THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET A

Chapters 1–4: **Orienting Yourself: The Use of Coordinates, Introduction to Linear Polynomials, The World of Numbers, Exploring Algebraic Identities** | Class 9 Maths | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 12 × 1 = 12

Q.	Correct Option	Brief Reasoning
1	**(c)** 13 units	$\sqrt{(4-(-1))^2+(11-(-1))^2}=\sqrt{25+144}=13$
2	**(c)** Quadrant IV	signs of (x, y) are $(+, -)$
3	**(d)** $(2,\ 3)$	$\left(\frac{5+(-1)}{2}, \frac{9+(-3)}{2}\right)$
4	**(a)** 3	highest power of the variable is 3
5	**(b)** $-3n+13$	common difference -3, so $a_n=-3n+13$
6	**(a)** falls (negative slope)	slope $-2<0$
7	**(b)** non-terminating repeating	denominator 6 has a prime factor other than 2 and 5
8	**(b)** Hippasus	as per the chapter
9	**(d)** rational	$=9$
10	**(d)** $9x^2+24x+16$	$(a+b)^2=a^2+2ab+b^2$
11	**(c)** $49x^2-4$	$(a+b)(a-b)=a^2-b^2$
12	**(d)** $(2s+5t)^2$	identify $a^2+2ab+b^2=(a+b)^2$

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
13	$PA^2=1+9=10$; $PB^2=9+1=10$. Yes, P is equidistant.	1+1
14	At $x=3$: $4(3)+(-1)=11$; at $x=0$: $4(0)+(-1)=-1$	1+1
15	$a_n=3n+(1)$; $a_{10}=3(10)+(1)=\mathbf{31}$	1+1
16	9 has a prime factor other than 2 and 5 ⇒ non-terminating repeating	1+1
17	$\frac{25p^2}{4}+\frac{15pq}{2}+\frac{9q^2}{4}$	2

SECTION — C (Short Answer) — Answers — 6 × 3 = 18

Q.	Expected Answer / Value Points	Marks
18	$AB=5=DC$, $BC=3=AD$ (opposite sides equal). Diagonals $AC=BD=\sqrt{25+9}=\sqrt{34}$ (equal). A parallelogram with equal diagonals is a rectangle.	1+1+1
19	$C(15)=100+900=$ ₹1000. $100+60d=700\Rightarrow d=10$ km. Positive slope 60 ⇒ linear growth.	1+1+1
20	$x=2.45\overline{37}$. $100x=245.\overline{37}$; $10000x=24537.\overline{37}$. $9900x=24292\Rightarrow x=\frac{24292}{9900}=\frac{6073}{2475}$.	1+1+1
21	$(2n)^3-3(2n)^2(5m)+3(2n)(5m)^2-(5m)^3=(2n-5m)^3$.	3
22	Write as $(3x)^2+(2y)^2+z^2+2(3x)(2y)+2(3x)(z)+2(2y)(z)=(3x+2y+z)^2$. Answer: $(3x+2y+z)^2$	3
	OR: $p^4-16=(p^2-4)(p^2+4)=(p-2)(p+2)(p^2+4)$; denominator $=(p-2)^2$. Result: $\frac{(p+2)(p^2+4)}{p-2}$.	1+1+1
23	$MA=\sqrt{9+16}=5$, $AG=\sqrt{36+64}=10$, $MG=\sqrt{81+144}=15$. $MA+AG=MG$ ⇒ collinear.	1+1+1
	OR: $AB=\sqrt{36+0}=6$; $AC=\sqrt{9+16}=5$; $BC=\sqrt{9+16}=5$. Two sides ($AC=BC=5$) are equal, so the triangle is isosceles.	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
24	(i) ₹7,600. (ii) 10000, 9200, 8400, 7600, 6800, 6000, 5200, 4400, 3600. (iii) $v=10000-800t$; constant decrease ₹800 per year. (iv) $10000-800t=2000\Rightarrow t=10$ years.	1+1+2+1
	OR: (i) $2n-1$. (ii) 29 and 51. (iii) $2n-1=21\Rightarrow n=11$; $n=24$ for 47. (iv) Slope 2 = constant difference between consecutive terms.	1+1+2+1
25	(i) $x^2+8x+15=(x+3)(x+5)$. (ii) $x(x-4)=96\Rightarrow(x-12)(x+8)=0$, so $x=12$ m, breadth 8 m. (iii) $2(12+8)=40$ m.	1+3+1

	OR: (i) $(3a - b + 2c)^2$ [check: $9a^2 + b^2 + 4c^2 - 6ab + 12ac - 4bc$ ✓]. (ii) $(4s - 5t)^2$.	3+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET B

Chapters 1–4: **Orienting Yourself: The Use of Coordinates, Introduction to Linear Polynomials, The World of Numbers, Exploring Algebraic Identities** | Class 9 Maths | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 12 × 1 = 12

Q.	Correct Option	Brief Reasoning
1	**(a)** $(8, -1)$	reflection in the y-axis changes the sign of the x-coordinate
2	**(d)** 10 units	$\sqrt{(6-(0))^2+(8-(0))^2} = \sqrt{36+64} = 10$
3	**(c)** $(1, 5)$ and $(5, 5)$	shift vertically by 3
4	**(b)** does not lie on the line	RHS at $x = 1$ is 7, not equal to 8
5	**(d)** 5	substitute $x = 2$ to get 5
6	**(b)** −10	term without the variable
7	**(d)** 2	number of places $= \max(a, b)$ in 2^a5^b
8	**(b)** $\frac{5}{11}$	standard method ⇒ $\frac{5}{11}$
9	**(c)** $\frac{5}{8}$	$\frac{1}{2}\left(\frac{1}{2}+\frac{3}{4}\right) = \frac{5}{8}$
10	**(a)** $4x^2 + 20x + 25$	$(a+b)^2 = a^2 + 2ab + b^2$
11	**(a)** $x^3 + 6x^2 + 12x + 8$	$(a+b)^3 = a^3 + 3a^2b + 3ab^2 + b^3$
12	**(b)** $(3x + 4)^2$	identify $a^2 + 2ab + b^2 = (a+b)^2$

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
13	Midpoint $= \left(\frac{-5+(1)}{2}, \frac{3+(9)}{2}\right) = (-\mathbf{2}, \mathbf{6})$	1+1
14	At $x = 1$: $-2(1) + (6) = 4$; at $x = 4$: $-2(4) + (6) = -2$	1+1
15	8 has prime factors only 2 and/or 5 (2^a5^b) ⇒ terminating	1+1
16	$4 - 15 = -11$ °C.	1+1
17	$9s^2 - 24st + 16t^2$	2

SECTION — C (Short Answer) — Answers — 6 × 3 = 18

Q.	Expected Answer / Value Points	Marks
18	Sum of coordinates: if A, B, C are vertices with D mid of BC etc., then $A = E + F - D = (1, 7)$, $B = D + F - E = (-1, -1)$, $C = D + E - F = (11, 3)$. Check: midpoint of $BC = (5, 1) = D$ ✓	1+1+1
19	$v = 10000 - 800t$; $v(3) = 7600$. $10000 - 800t = 4000 \Rightarrow t = 7.5$ years.	1+1+1
20	If $3 + \sqrt{2} = r$ (rational), then $\sqrt{2} = r - 3$ would be rational — contradiction. Hence $3 + \sqrt{2}$ is irrational.	1+1+1
21	$(x + y + z)^2 = 100 = 38 + 2(xy + yz + zx) \Rightarrow xy + yz + zx = 31$.	1+1+1
22	$AB = \sqrt{36+0} = 6$; $AC = \sqrt{9+16} = 5$; $BC = \sqrt{9+16} = 5$. Two sides ($AC = BC = 5$) are equal, so the triangle is isosceles.	1+1+1
	OR: $AB = 4 = DC$, $BC = 3 = AD$ (opposite sides equal). Diagonals $AC = BD = \sqrt{16+9} = 5$ (equal). A parallelogram with equal diagonals is a rectangle.	1+1+1
23	$800 = 10a + b$, $1100 = 15a + b \Rightarrow a = 60,\ b = 200$. $y = 60x + 200$; for $x = 20$: ₹1400.	2+1
	OR: $0 = 32a + b$, $100 = 212a + b \Rightarrow 180a = 100$, $a = \frac{5}{9}$, $b = -\frac{160}{9}$. °C $= \frac{5}{9}(^\circ F - 32)$. At 98.6 °F: $\frac{5}{9} \times 66.6 = 37$ °C.	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
24	(i) $AB = 8$, $AC = BC = \sqrt{16+36} = 2\sqrt{13}$ ⇒ isosceles. (ii) $D(6, 3)$, $E(2, 3)$, $F(4, 0)$; $DE = 4$, $EF = \sqrt{4+9} = \sqrt{13}$, $FD = \sqrt{4+9} = \sqrt{13}$. Perimeter $= 4 + 2\sqrt{13}$ (half of that of ABC).	2+3
	OR: (i) $PQ = \sqrt{36+64} = 10$. (ii) $M = (1, -1)$. $MP = \sqrt{9+16} = 5$, $MQ = \sqrt{9+16} = 5$. Each $= \frac{1}{2}PQ$ ✓	2+3
25	(i) Contradiction proof: $\sqrt{2} = \frac{p}{q}$, $2q^2 = p^2$, p and q both even — contradiction (3). (ii) If $3\sqrt{2} - 5 = r$ (rational) then $\sqrt{2} = \frac{r+5}{3}$ rational — contradiction (2).	3+2

	OR: (i) All are primes between 10 and 20; next primes 23, 29, 31. (ii) $\frac{15}{2} \times 12 = 90$ ingots. (iii) No: $3 - 5 = -2$ is not a natural number.	2+2+1

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ANSWER KEY & MARKING SCHEME SET C

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SECTION — A (Objective Type) — Answers — 12 × 1 = 12

Q.	Correct Option	Brief Reasoning
1	**(c)** y-axis (measured along the x-axis)	definition of abscissa
2	**(b)** $(-2,\ 3)$	$\left(\frac{-6+(2)}{2}, \frac{-2+(8)}{2}\right)$
3	**(a)** $a = b$	the coordinates are equal in order only when $a = b$
4	**(b)** the origin	$x = 0 \Rightarrow y = 0$
5	**(b)** 2	highest power of the variable is 2
6	**(c)** -1 and -2	compare with $y = ax + b$
7	**(d)** 56	Brahmagupta's rules for signs
8	**(c)** non-terminating repeating	denominator 14 has a prime factor other than 2 and 5
9	**(b)** −7	−7 is an integer
10	**(b)** $49x^2 + 14x + 1$	$(a+b)^2 = a^2 + 2ab + b^2$
11	**(c)** $(x + 5)\,(x + 6)$	find a, b with $a + b$ = middle coefficient and ab = constant
12	**(c)** 841	$29^2 = (30 - 1)^2$

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
13	$AB = \sqrt{(12-(0))^2 + (0-(5))^2} = \sqrt{144+25} = \sqrt{169} = \mathbf{13}$ units	1+1
14	At $x = 0$: $5(0) + (-3) = -3$; at $x = -1$: $5(-1) + (-3) = -8$	1+1
15	Let $x = 0.1\overline{3}$. $10x = 1.\overline{3}$ and $100x = 13.\overline{3}$; subtracting, $90x = 12 \Rightarrow x = \frac{2}{15}$	2
16	$x^2 + xy + \frac{y^2}{4}$	2
17	$(s - 8)\,(s - 3)$	2

SECTION — C (Short Answer) — Answers — 6 × 3 = 18

Q.	Expected Answer / Value Points	Marks
18	$AB = \sqrt{36+0} = 6$; $AC = \sqrt{9+16} = 5$; $BC = \sqrt{9+16} = 5$. Two sides ($AC = BC = 5$) are equal, so the triangle is isosceles.	1+1+1
19	₹5 coins $= x$, ₹2 coins $= 3x$. $5x + 6x = 88 \Rightarrow x = 8$. So 8 five-rupee coins and 24 two-rupee coins.	1+1+1
20	Assume $\sqrt{3} = \frac{p}{q}$ (co-prime). $3q^2 = p^2 \Rightarrow 3 \mid p$; $p = 3k \Rightarrow q^2 = 3k^2 \Rightarrow 3 \mid q$. Both divisible by 3 — contradiction. So $\sqrt{3}$ is irrational.	1+1+1
21	$(p)^3 + 3p^2(2q) + 3p(2q)^2 + (2q)^3 = (p + 2q)^3$, so the side is $p + 2q$.	3
22	$x + 4x = 300 \Rightarrow x = 60$. Pieces: 60 ft and 240 ft.	1+1+1
	OR: $400 = 10a + b$, $500 = 14a + b$. Subtract: $4a = 100 \Rightarrow a = 25$; $b = 400 - 250 = 150$. $y = 25x + 150$.	1+1+1
23	Write $3 = \frac{18}{6}$ and $4 = \frac{24}{6}$: $\frac{19}{6}, \frac{20}{6}, \frac{21}{6}, \frac{22}{6}, \frac{23}{6}$ (any five valid).	1+1+1
	OR: $100x = 45.\overline{45}$, $99x = 45 \Rightarrow x = \frac{45}{99} = \frac{5}{11}$. Check: $5 \div 11 = 0.4545\ldots$ ✓	2+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
24	(i) e.g. $y = 2x - 1$: $(0, -1), (1, 1)$; $y = 2x + 1$: $(0, 1), (1, 3)$; $y = 2x + 5$: $(0, 5), (1, 7)$. (ii) All parallel (same slope 2); they shift up/down as b changes. (iii) $(0, 5)$.	3+1+1
	OR: (i) t: 0,1,2,3,4,5,6 → h: 3, 2.5, 2, 1.5, 1, 0.5, 0. (ii) Linear decay — constant decrease of 0.5 m per month. (iii) At $t = 6$ months. (iv) Straight line through $(0, 3)$ and $(6, 0)$; slope -0.5, h-intercept 3.	1+1+1+2
25	(i) $25 + 49 - 72 = 2$. (ii) $(n-1)^2 + (n+1)^2 - 2n^2 = 2$ (2). (iii) $(n-2)^2 + (n+2)^2 - 2n^2 = 8$ — a constant 8, not 2.	1+2+2
	OR: (i) $100 = 38 + 2(xy + yz + zx) \Rightarrow 31$. (ii) $x^3 + y^3 + z^3 - 3xyz = (10)(38 - 31) = 70 \Rightarrow x^3 + y^3 + z^3 = 70 + 75 = 145$.	2+3

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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